VISTARA-X: Simple Bhasha Mein

Ek line mein: VISTARA-X ek robot hai jo computer ke andar bane ek warehouse (simulation) mein chalta hai. Ye khud apne aas-paas ka map banata hai, tumhari baat (Hindi ya English, bolkar ya likhkar) samajhta hai, target tak safe raasta dhoondta hai, raasta block ho jaaye toh ruk jaata hai, aur naya raasta bana leta hai. Ise bina sune, sirf padhkar aur type karke bhi poora chalaya ja sakta hai.

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1. Idea kya hai

Problem

Socho ek delivery robot hai jo warehouse mein point A se point B tak jaata hai. Raaste mein:

- shelves aur pallets hain jo kabhi hilte nahi,
- boxes hain jo koi bhi achaanak raaste mein rakh sakta hai,
- log hain jo idhar-udhar chal rahe hain.

Robot ko target tak **bina kisi cheez se takraaye** pahunchna hai. Aur agar sach mein koi raasta hi nahi hai, toh use **imaandari se bolna** chahiye ki "raasta nahi hai". Na crash kare, na jhooth-mooth ki koshish ka dikhawa kare.

Humara solution

Humne robot ka "dimaag" banaya aur use asli hardware ki jagah ek realistic simulation mein test kiya. Robot ek simple loop baar-baar chalata hai, ek second mein kai baar:

```
Suno → Dekho → Map banao → Raasta plan karo → Chalo → Check karo → Block hua toh naya raasta
```

Bilkul waise hi jaise tum bheed waale kamre mein chalte ho. Pehle dekhte ho, raasta chunte ho, chalna shuru karte ho. Koi saamne aa jaaye toh rukte ho aur doosra raasta le lete ho.

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2. Ye kaam kaise karta hai (step by step)

Step	Robot kya karta hai	Real life mein jaise
1. Dekhna	Sensors padhta hai: camera, depth camera, LiDAR (laser se doori naapne waala scanner), ultrasonic sensors, IMU (ghoomne ka sensor) aur wheel counters	Tumhari aankhein aur balance
2. Pehchaanna	YOLO AI model camera ki image dekh kar batata hai ki saamne kya hai (jaise "person")	Saamne khade insaan ko pehchaan lena
3. Samajhna	Floor ka map banata hai: 20 cm ke chhote-chhote squares ka grid, har square ya toh free hai ya blocked. Blocked squares ke around ek extra safety border bhi lagta hai	Ye pata hona ki furniture kahan rakha hai

Step	Robot kya karta hai	Real life mein jaise
4. Plan karna	**A*** (A-star) algorithm grid par sabse chhota safe raasta dhoondta hai. Use karne se pehle raasta dobara check hota hai	Google Maps ka route dhoondna
5. Chalna	Controller robot ko raaste par smoothly chalata hai	Raaste par chalna
6. Check aur re-plan	Har pal agle 4 metre check karta hai. Kuch raasta block kare toh robot **ruk jaata hai**, 0.3 second rukta hai, naya raasta banata hai, check karta hai, phir chalta hai	Koi saamne aa jaaye toh side se nikal jaana

Safety: teen alag-alag checks

Robot tabhi chalta rehta hai jab teeno checks bolein ki sab safe hai:

1. **Raasta check:** kya plan kiya hua raasta map par abhi bhi khaali hai?
2. **Doori waale sensors:** LiDAR aur ultrasonic naapte hain ki cheezein kitni paas hain. 0.9 m se kam doori par robot dheere ho jaata hai, 0.3 m se kam par ruk jaata hai.
3. **Final safety lock:** physics check aisi koi bhi movement rok deta hai jisse robot kisi cheez ke 5 cm se zyada paas aa jaaye.

Agar koi bhi safe raasta nahi hai, toh robot **Blocked** dikhata hai aur apni jagah se hilta nahi.

Map banana (SLAM)

Robot ko pehle se map dena zaroori nahi. Woh khud ghoomkar map bana leta hai:

1. **Andaaza:** wheel counters batate hain ki robot kitna chala. Ye thoda galat hota jaata hai, kyunki wheels fisalte hain.
2. **Sudhaar:** har LiDAR scan ko ab tak bane map se milaya jaata hai. Jahan scan sabse achha baithta hai, wahi robot ki sahi jagah maani jaati hai.
3. **Map:** har laser ray jahan se guzri woh jagah "khaali", jahan takraayi woh "deewar", aur jahan koi ray nahi pahunchi woh "anjaan".

Asar: ek lambe chakkar mein sirf wheel counters ki galti **140 cm** tak pahunch gayi, lekin SLAM ki galti **8–10 cm** hi rahi. **Explore and map automatically** dabao toh robot khud anjaan jagahon ki taraf jaakar ~30 second (simulated time) mein warehouse ka 83–90% map bana leta hai, bina takraaye.

Map ko ROS waale format (PGM + YAML) mein save, load, download aur import bhi kar sakte ho. Ek tayyar map `warehouse_demo` repo mein hai.

Baat samajhna (voice aur text commands)

- **Bolo ya likho:** "Robot ko reception tak le jao", "रोबोट को चार्जिंग डॉक तक ले जाओ", "Take the robot to the storage room", "रुको", "थोड़ा रुको", "फिर से चलो", "वापस जाओ".
- **Speech:** Whisper naam ka AI model tumhare apne computer (backend) par chalta hai; awaaz internet par nahi jaati. Ek command mein ~3–6 second lagte hain.
- **Samajhna fixed niyamon se hota hai, AI se nahi.** Isliye robot kabhi andaaze se jagah nahi chunta. Anjaan jagah ("cafeteria") boli toh woh puchhta hai: "Ye jagah mujhe nahi pata; ye jagahein pata hain: ...".

- **Pehle dikhao, phir chalo:** robot batata hai ki usne kya samjha (bhasha, jagah, raasta kitne metre). Tum **Confirm** dabaoge tabhi woh chalega.
- **Emergency stop** ek alag bada button hai. Ye bolne ya samajhne par depend nahi karta aur hamesha kaam karta hai. Command mein kahin bhi "stop/ruko" ho toh robot turant rukta hai.
- **Jawab usi bhasha mein:** Hindi mein bologe toh jawab Hindi mein, Hinglish mein bologe toh Hinglish mein. Chaho toh jawab bolkar bhi sunaya jaata hai, lekin text hamesha screen par rehta hai.

Bina sune istemaal (accessibility)

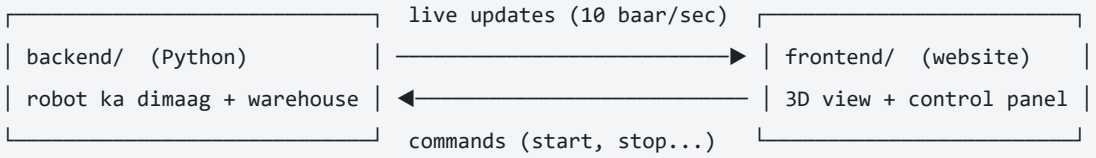
Jo log sun nahi sakte ya kam sunte hain, woh bhi sab kuch chala sakte hain:

- Har command type karke ya list se jagah chunkar di ja sakti hai, aur saari baatcheet (transcript) screen par rehti hai.
- Har zaroori warning (raasta block, Blocked, takkar, error, pahunch gaya) bade banner mein **icon aur text** ke saath aati hai, sirf rang ya awaaz se nahi.
- **Accessible mode:** bada text, tez contrast, aur alerts tab tak rehte hain jab tak tum khud band na karo.
- Poora flow sirf keyboard se chalaya ja sakta hai.

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3. Ye bana kaise hai

Project ke do hisse hain jo aapas mein baat karte hain:



Backend (dimaag), `backend/` folder mein

Hissa	Kya karta hai	Kahan hai
Simulation	Virtual warehouse ko 1 second mein 20 baar update karta hai: robot, shelves, boxes, log	`app/simulation/`
Path planning	Grid map par A* search	`app/navigation/`
Steering	Robot ko raaste par chalata hai	`app/control/`
Sensors	Simulated camera, LiDAR, ultrasonic, IMU, wheel counters (asli jaisa thoda noise bhi)	`app/sensors/`
AI detection	YOLOv8 models jo camera image mein cheezein pehchaante hain	`app/perception/`
Mapping (SLAM)	LiDAR se map banana, robot ki jagah sudhaarna, khud explore karna	`app/mapping/`

Hissa	Kya karta hai	Kahan hai
Commands aur speech	Hindi/English command samajhna, Whisper se awaaz ko text banana	`app/commands/`
Physics	PyBullet takkar check karta hai aur camera ki image banata hai	`app/simulation/physics.py`
API	Website se commands leta hai aur live data bhejta hai	`app/api/`

Isme use hua: **Python, FastAPI, OpenCV, NumPy, SciPy, PyTorch, Ultralytics YOLO, PyBullet, faster-whisper**.

Frontend (screen), `frontend/` folder mein

Hissa	Kya dikhata hai
3D view	Warehouse, robot, uska raasta, LiDAR ke dots aur log, sab 3D mein (2D flat map bhi hai)
Mission controls	Target chuno, Start, Pause, Stop, obstacles ya log add karo
Commands	Text box, mic button, jagah chunne ki list, Confirm/Cancel, Emergency stop, transcript history
Mapping	Live map, robot ka andaaza vs asli jagah, LiDAR scan, map save/load/import
Alerts	Bade visual alerts aur Accessible mode ki settings
Panels	Sensors, AI detection, navigation map, live numbers, event log, system health
Demos	Ek click waala guided demo aur blocked-route demo
Challenge	"Trap the robot" game: 5 barriers se robot ko rokne ki koshish karo

Isme use hua: **React, Vite, Three.js (3D), Framer Motion (animations)**.

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4. Use kaise karein

Chalana kaise hai (Windows)

Terminal 1, backend:

```
cd backend
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
pip install -r requirements-optional.txt
python -m uvicorn app.main:app --port 8000
```

Terminal 2, website:

```
cd frontend
npm install
npm run dev
```

Browser mein **http://localhost:5173** kholo aur **Live mission** par click karo.

Try karke dekho

1. **Laptop slow hai?** Upar **Demo mode** tick karo. 3D view halka ho jaata hai aur robot dheere chalta hai, toh follow karna aasaan hai. Demo se pehle ek baar **Speak a command** daba do, taaki Whisper model load ho jaaye.
2. **Map banao:** Mapping panel mein **Explore and map automatically** dabao. Robot khud ghoomkar map banayega. Ya tayyar `warehouse_demo` map **Load** karo.
3. **Bolkar ya likhkar bhejo:** Commands panel mein "Robot ko reception tak le jao" type karo, ya mic dabakar bolo. Card mein dekho ki robot ne kya samjha, phir **Confirm and go** dabao.
4. **Ya click karke target chuno: Select destination** dabao aur floor par kahin bhi click karo, ya `18.5` aur `6` type karke **Start mission** dabao.
5. **Raasta block karo:** robot jab shuruaat mein hi ho, tab **Drop obstacle on route** dabao. Robot rukega, **Replanning** dikhayega, aur obstacle ke around se nikal jaayega.
6. **Finish:** robot target tak pahunch jaata hai, aur "Destination reached" ka alert aata hai. Collision count 0 hi rehta hai.
7. **Anjaan ya band jagah:** "Take the robot to the cafeteria" likho; robot puchhega ki kaunsi jagah. Phir **Run blocked-route demo** dabao: target chaaron taraf deewaron se band hai, toh robot **Blocked** dikhata hai aur hilta nahi.
8. **Bina awaaz ke: Display and alert settings** mein **Accessible mode** tick karo, aur upar ke steps sirf keyboard aur screen padhkar karo.
9. **Game khelo: Trap the robot** panel mein **Start challenge** dabao. Tumhare paas 5 barriers hain. Robot chal raha hota hai aur tumhe floor par click karke uska raasta rokna hai. Thoda bhi gap chhoda toh robot naya raasta dhoondh kar nikal jaayega (robot jeeta). Poora band kar diya toh robot imaandari se **Blocked** bolega (tum jeete). Dono case mein collisions 0 hi rehte hain. Hint: x = 15 par dono racks ke beech ka gap dekho.
10. **Aur khelo:** boxes ya chalta hua insaan add karo, LiDAR band karo, YOLO AI model chuno, ya camera view badlo (Orbit, Top, Follow robot).

Kabhi bhi kuch gadbad lage toh **Emergency stop** dabao.

Real duniya mein iska kya use hai

- **Warehouses aur factories:** shelves ke beech saamaan dhone waale robots.
- **Hospitals aur clinics:** bheed waale corridors mein dawaai pahunchane waale robots, jinhe staff bolkar jagah bata sake.
- **University campus aur offices:** visitors Hindi ya English mein bolkar robot ko kahin bhej sakein.
- **Deaf aur kam sunne waale log:** bina awaaz ke, sirf text aur visual alerts se robot chalana.
- **Delivery aur cleaning robots:** logon ke beech safely chalna.

- **Banane se pehle testing:** mehenga hardware khareedne se pehle robot ka dimaag simulation mein test kar sakte hain.

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5. Imaandari se kuch baatein

- Abhi sab kuch **simulated** hai. Koi asli robot ya sensor connected nahi hai.
- AI models aam photos (COCO dataset) par trained hain, isliye humare computer se bane warehouse mein kam cheezein pehchaante hain.
- Map bhi simulated LiDAR se banta hai. Asli sensors waala ROS 2 SLAM is laptop par install nahi hai, isliye woh chalaya nahi gaya (README mein uska setup guide hai).
- Bolkar command English mein computer ki banayi awaaz (TTS) se test hui hai. Hindi bolkar command ka automatic test nahi ho paaya, kyunki laptop mein Hindi awaaz ya recording nahi thi. Hindi likhkar di gayi commands tested hain.
- Robot sirf jaani-pehchaani jagahein aur bataye gaye tarah ke vaakya samajhta hai, har tarah ki baat nahi.
- Sign language samajhna abhi nahi bana hai.
- Ye saare real-world use sirf intended hain; kisi asli building mein test nahi hue.
- Humne automated tests chalaye hain (138 backend, 32 browser) aur SLAM ki galti simulation mein naapi hai, lekin bahut saare runs par success rate abhi naapa nahi hai.

Poori technical detail ke liye README.md dekho.

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